MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Quick Reference Guideline for Ambulatory Treatment of Urinary Tract Infections in Adults

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Asymptomatic Bacteriuria	Uncomplicated Lower Tract UTI (Cystitis, infection limited to the bladder)
Prostatitis	Complicated UTI (infection beyond the bladder)

New classifications of uUTI and cUTI

Old Classifications

Uncomplicated UTI: Acute cystitis in afebrile nonpregnant premenopausal women with no diabetes and no urologic abnormalities



Acute Pyelonephritis: Acute kidney infection in women otherwise meeting the definition of uncomplicated UTI above



Complicated UTI: All other UTIs

New Classifications

Uncomplicated UTI: Infection confined to the bladder in afebrile women or men

Complicated UTI: infection beyond the bladder in women or men

- Pyelonephritis
- Febrile or bacteremic UTI
- Catheter-associated (CAUTI)
- Prostatitis* (*not covered by these guidelines)



2025 IDSA Complicated UTI guideline updates

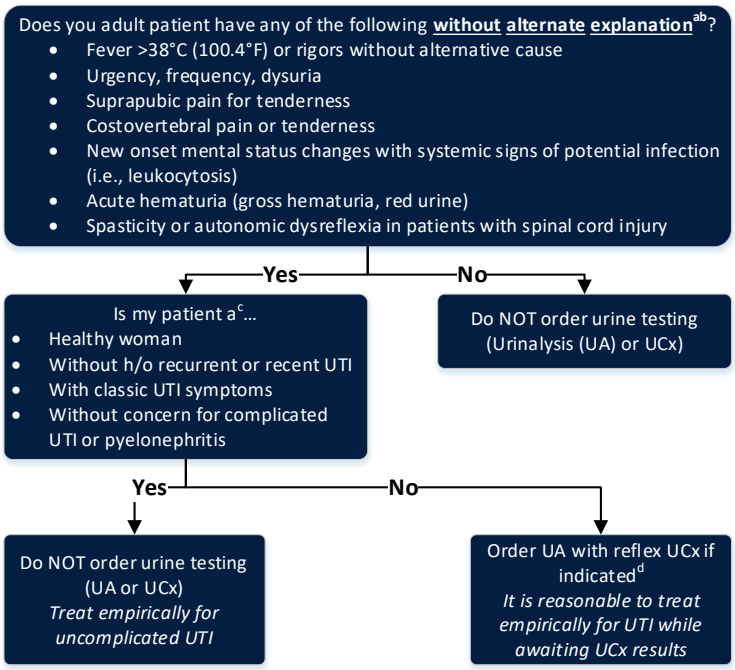
Classification	Drug/duration	Comments
Uncomplicated UTI (bladder only) • Most outpatients • Some hospitalized	Short durations: **Nitrofurantoin** x5 days **TMP-SMX** x3 days **Beta-lactams** x 5-7 days	Beta-lactams • generally considered inferior • Appropriate for: ○ Allergies, significant CKD, or resistance profiles • Require longer durations (5-7 days)
Complicated UTI (beyond the bladder) • Antibiotics required to have good tissue penetration	Standard duration: **TMP-SMX** x7 days **Fluoroquinolones** x5-7 days **Beta-lactams** x7 days	Beta-lactams • generally considered inferior • Appropriate for: ○ Allergies, significant CKD, or resistance profiles • Require longer durations (7 days)

UTI DIAGNOSTICS KEY POINTS:

- **Mental status changes in geriatric patients SHOULD NOT be worked up/treated for a UTI unless leukocytosis, hypotension, or ≥ 2 SIRS criteria are present.**
- The presence of pyuria, nitrites or positive urine cultures are NOT diagnostic for a UTI. A urinalysis or urine culture should only be utilized to **RULE OUT a UTI**, not rule in.
- **Except in pregnancy and prior to urologic procedures,** bacteria found in a urine culture without urinary specific symptoms or symptoms of systemic infection (Asymptomatic Bacteriuria) should not be treated, regardless of presence of nitrites or pyuria, bacterial density (CFU's), or isolation of resistant organisms.

When to Order a Urinalysis (UA) and Urine Culture (UCx): Asymptomatic bacteriuria (ASB) is often treated unnecessarily, accounting for a substantial burden of unnecessary antimicrobial use. National guidelines recommend against testing for ASB, except in select circumstances[1]. Urinary tract infections are over-diagnosed. Patient histories of symptoms that have elicited antibiotic treatment in the past for a "UTI" may or may not have been indicative of a true infection. As such, the following approach is recommended in all patients.

In the absence of signs or symptoms (see below) attributable to a urinary tract infection (UTI), patients with a positive UCx and/or pyuria on UA should not be treated with antibiotics irrespective of high bacterial colony count, or a multi-drug resistant organism. Therefore urine testing should only be obtained for appropriate reasons. A negative UA makes a UTI very unlikely to be the cause of the patient's symptoms, but a positive UA does not diagnose a UTI. *Urinary symptoms are needed to diagnose a UTI.* The following is an effective strategy for how and when to order a UA and/or UCx. In addition, depending on the clinical scenario, you should rule out the possibility of a sexually transmitted infection or vaginitis.



[a] Exceptions to this recommendation include patients that are pregnant or undergoing a urologic procedure. Clinical judgement should be used for patients with baseline cogntive/functional impairment presenting with new functional decline or falls with systemic signs of potenital infection (i.e., leukocytosis) and without an alternative etiology

[b] These ambulatory guidelines do not apply to severe sepsis, or patients with more severe presentations of illness, including hypotension, or ≥ 2 SIRS criteria (SIRS Criteria: Heart rate > 90 bpm, respiratory rate > 20 breaths per minute, temperature < 36°C (96.8°F), white blood count < 4,000 cells/mm^3, temperature > 38°C (100.4°F), white blood count > 12,000 cells/mm^3).

[c] In healthy women with classic signs and symptoms of a UTI, urine testing (UA or UCx) are not necessary, and the patient may be treated empirically. However, patients at risk for drug-resistant bacteria, patients with underlying health conditions putting them at risk for more serious illness, if suspicion for upper tract UTI is present (fever, flank pain), or patients with recurrent UTIs should have a UA and UCx sent. In addition, a negative UCx does not rule out a UTI in a patient with classic symptoms. Use clinical judgement and patient response to determine if antibiotics should be continued.

[d] A UCx will only be performed if a UA result indicates an inflammatory response, and therefore possible infection. Notable UA results include: detectable nitrites, leukocyte esterase, and bacteria. This progression is a strategy to decrease unnecessary antibiotic treatment in samples indicative of colonization and not infection.[2]

Clinical Setting	Empiric Therapy (Should take recent previous cultures into account)	Duration	Comments
Asymptomatic Bacteriuria[1] No symptoms of UTI (listed below), without alternative explanation: - Urgency, frequency, dysuria - Suprapubic pain or tenderness - Costovertebral pain or tenderness - Spasticity or autonomic dysreflexia in patients with spinal cord injury - Fever > 38°C or rigors without alternative cause - New onset mental status changes with systemic sign of potential infections (leukocytosis) - Acute hematuria without alternative explanation	In most circumstances, asymptomatic bacteriuria **should not be treated,** regardless of pyuria, bacterial density, or isolation of resistant organisms. Treatment is recommended in the following circumstances: - Pregnancy - Prior to urologic procedures <u>Treatment of Asymptomatic Bacteriuria in PREGNANCY</u> <u>Preferred:</u> **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min)** <u>Alternatives:</u> **Cephalexin*** 500 mg PO BID (avoid if high-risk allergy to penicillin or cephalosporin, see BLAES guidance) OR **Fosfomycin** 3 g PO x1 dose (less preferred due to cost) ***Adjust dose based on renal function **The Beers Criteria recommends avoiding use in geriatric patients > 65 years with a CrCl < 30 mL/min	Treatment of Asymptomatic Bacteriuria in PREGNANCY **Nitrofurantoin**: 5 days **Cephalexin**: 7 days **Fosfomycin**: 1 dose	• Surgical prophylaxis guidelines provide recommendations on antimicrobial prophylaxis prior to genitourinary operation • Available evidence does not support screening for, and treatment of, asymptomatic bacteriuria prior to implantation of prosthetic orthopedic[3], cardiac devices, or neurosurgical procedures/devices. • Fosfomycin should only be used for *E. coli* or *Enterococcus faecalis*. It may be considered on a case-by-case basis for other organisms depending on the severity of symptoms, allergies, resistance profile, and remaining oral options **Pregnancy**: • Urine culture should be sent and treatment adjusted based on susceptibilities. Follow-up urine cultures should be obtained for test of cure. • <u>Contraindicated throughout pregnancy</u>: ○ Fluoroquinolones and doxycycline • <u>Avoid in first 8 weeks of pregnancy</u>: ○ TMP-SMX

Uncomplicated Lower Tract Urinary Tract Infection (Cystitis)[4] Infection limited to the bladder in male or female patients not meeting criteria for complicated UTI or CA-UTI (see below)	Preferred: **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min)** 1st alternatives: **TMP-SMX*** 1 DS tab PO BID 2nd alternative: **Fosfomycin** 3 g PO x1 dose (consider cost) OR **Cephalexin*** 1000 mg PO q12h 3rd alternative: **Ciprofloxacin*** 500 mg PO BID OR **Levofloxacin*** 500 mg PO daily 4th alternative: **Gentamicin*** 5 mg/kg IV/IM x1 OR **Tobramycin*** 5 mg/kg IV/IM x1 OR **Amikacin*** 15 mg/kg IV/IM x1 (dose for ideal body weight) *Adjust dose based on renal function **The Beers Criteria recommends avoiding use in geriatric patients > 65 years with a CrCl < 30 mL/min	**Nitrofurantoin**: 5 days **TMP-SMX**: 3 days **Ciprofloxacin / levofloxacin**: 3 days **Fosfomycin**: 1 dose **Cephalexin**: 5-7 days For patients with recurrent or relapsed UTIs, or slow/delayed response to treatment, use clinician judgement for duration, as longer durations may be appropriate. In pregnancy, treatment durations should be 5-7 days.	• Fosfomycin is less effective than nitrofurantoin and should only be used if a contraindication to nitrofurantoin exists • Fosfomycin should only be used for *E. coli* or *Enterococcus faecalis*. It may be considered on a case-by-case basis for other organisms depending on the severity of symptoms, allergies, resistance profile, and remaining oral options • Fluoroquinolones are not recommended as 1st-line agents due to high rates of *E. coli* resistance and propensity for collateral damage (resistance, *C. difficile* infection)[4,5,6]. Use should be reserved when other options are not feasible • Oral beta-lactams are not recommended first line as they have a higher risk of treatment failure. • Single dose aminoglycoside may be a good option for patients with antibiotic-resistant cystitis with symptoms limited to the bladder in patients who do not otherwise have an indication for admission. This should be done in coordination with their Infectious Diseases provider and renal function should be taken into account. • Renal transplant patients are beyond the scope of this guideline. Careful consideration should be given to determine if a shorter antibiotic course is appropriate for the individual patient. **Pregnancy**: • Urine culture should be sent and treatment adjusted based on susceptibilities. Follow-up urine cultures should be obtained for test of cure. • Contraindicated throughout pregnancy: ○ Fluoroquinolones and doxycycline • Avoid in first 8 weeks of pregnancy: ○ TMP-SMX

Clinical Setting	Empiric Therapy (Should take recent previous cultures into account)	Duration	Comments
Catheter-associated UTI (CAUTI): without symptoms indicating infection beyond the bladder Includes indwelling foley, suprapubic catheter, intermittent straight catheter (ISC)	Preferred **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min). 1st Alternative **TMP-SMX*** 1 DS tab PO BID 2nd Alternative **Cephalexin*** 1000 mg PO BID (avoid if anaphylaxis to penicillin) 3rd Alternative **Ciprofloxacin** 500 mg PO q12h OR **Levofloxacin** 500 mg PO q24h	7 days FQ: 5 days For patients with recurrent or relapsed UTIs, use clinician judgement for duration, as longer durations may be appropriate.	• Asymptomatic bacteriuria with or without pyuria in catheterized patients are NOT an indication for treatment • Remove urinary catheter whenever possible • If catheter removed and instigating symptoms have resolved, likely no infection was present and no further treatment is needed • Antimicrobial choice should be adjusted based on urine culture and susceptibility testing • Oral beta-lactams are not recommended first line as they have a higher risk of treatment failure.

Clinical Setting	Empiric Therapy (Should take recent previous cultures into account)	Duration	Comments
Complicated Urinary Tract Infection (Infection beyond the bladder) Including: • Febrile UTI • Pyelonephritis	Preferred: **TMP-SMX*** 1 DS tab PO BID + **Ceftriaxone** 1 g IM x1 dose Alternatives: **Levofloxacin*** 750 mg PO daily (or **ciprofloxacin** 750 mg PO BID) + **Ceftriaxone** 1 g IM x 1 dose *Adjust dose based on renal function Ceftriaxone is added to empiric regimens due to high resistance rates to fluoroquinolones and TMP/SMX. Definitive therapy can be decided by culture results.	**TMP-SMX:** 7 days **FQ:** 5-7 days For patients with relapsed complicated UTI, delayed symptom resolution, or urinary diversion, recent urologic surgery, anatomic abnormalities, use clinician judgement for duration, as longer durations may be appropriate. Renal transplant patients with complicated UTI (with infection beyond the bladder) should be treated with a minimum 14-day duration. ID consult is recommended.	• **Urine culture and susceptibility testing should be obtained** • Patients with UTI with severe sepsis or bacteremia or obstruction or pregnant with infection beyond the bladder should be admitted (see inpatient guidelines). For patients with febrile UTI or pyelonephritis, consider admitting patients to be hospitalized if endorse persistently high fevers > 38.4C, unable to maintain oral hydration or take oral medications, have a history of resistance to oral options, suspected urinary tract obstruction or concerns regarding patient adherence • Nitrofurantoin and fosfomycin should be avoided in complicated UTI treatment given concern for upper-tract or parenchymal involvement • Oral beta-lactams carry a higher risk of treatment failure.

*Renal Dosing Recommendations
**The Beers Criteria recommends avoiding use in geriatric patients > 65 years with a CrCl < 30 mL/min

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Revision History

Date	**Revision**
07/2026	Added clarification comment on renal transplant duration for uncomplicated UTI

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source